

Primordial Black Hole Evaporation Threshold from the Projective Dynamic Logo Framework: A Falsifiable Prediction for Fermi-LAT

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Abstract

The Projective Dynamic Logo (PDL) framework derives all fundamental physical constants from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields [1, 2, 3]. A central result of the framework is that the effective gravitational coupling is density-dependent: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, where $\sigma(N) = 1 - (1 - \kappa_{\text{PDL}})^N$ and $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem of the axioms [4, 5]. At the epoch of primordial black hole (PBH) formation ($N = N_{\text{CMB}} = 40$, derived from the neutron quintuplet without cosmological input [6]), this coupling suppression extends PBH lifetimes by a factor $\sigma(40)^{-2} = 1.4007$. The resulting shift in the PBH survival mass threshold is $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 1.1189 M_{\text{GR}}^*$, corresponding to an upward shift of +11.89% relative to the standard general-relativistic value $M_{\text{GR}}^* \approx 5.1 \times 10^{14} \text{ g}$. This prediction implies an excess of Hawking radiation flux in the isotropic gamma-ray background (IGRB) at $E_{\text{peak}} \approx 90\text{--}105 \text{ MeV}$ from PBHs in the mass window $[5.10, 5.71] \times 10^{14} \text{ g}$ that are still active according to PDL but are predicted to have already evaporated by standard general relativity. The prediction is fully determined by a single unconditional theorem (κ_{PDL}) and contains no free parameter beyond the standard-model degrees of freedom entering the Hawking spectrum. We identify the precise observational test using the public codes BLACKHAWK [7] and ISATIS [8] together with the Fermi-LAT IGRB measurements of Ackermann et al. [9] and the constraint compilation of Carr et al. [10], and invite the community to carry it out.

Item	Status
$\kappa_{\text{PDL}} = 310\varphi/11017$	Unconditional theorem of C1–C4 [11, 5]
$N_{\text{CMB}} = 40$	Theorem (neutron quintuplet, no cosmological input) [6]
$\sigma(40) = 1 - (1 - \kappa_{\text{PDL}})^{40}$	Exact closed-form expression
$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^*$	Unconditional corollary [12]
Numerical value $M_{\text{PDL}}^* = 5.7063 \times 10^{14} \text{ g}$	Computed (this document, verified)
Shift +11.89%	Computed (this document, verified)
Excess flux at $\sim 100 \text{ MeV}$	Falsifiable prediction
Test via BLACKHAWK/ISATIS	To be performed by the community

Table 1: Epistemic status of the main claims in this document.

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1 Introduction

The Projective Dynamic Logo (PDL) framework [1, 13] derives fundamental physical constants from four axioms on finite signed graphs without presupposing any background spacetime, particles, or fields. The minimal stationary closure under these axioms is the complete graph K_4 on four vertices and six edges [14], identified with the electron prototype. The proton is the minimal hierarchical composite, characterised by the unique integer quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$ [15, 16].

A central structural result of the framework is the density-dependent effective gravitational coupling [17, 18, 4]:

$$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}, \quad \sigma(N) \equiv 1 - (1 - \kappa_{\text{PDL}})^N, \quad (1)$$

where N is the number of coherent closure units in the local environment, and $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem of the PDL axioms [11, 5]. Newton's gravitational constant G_{Newton} is recovered as the $N \rightarrow \infty$ limit $G_{\text{Newton}} = G_{\text{PDL}}$, whilst at any finite density the coupling is suppressed by $\sigma(N) < 1$.

This density dependence has direct consequences for primordial black holes (PBHs), which formed at early cosmological epochs when the local closure density was characterised by $N = N_{\text{CMB}} = 40$ [6, 19]. Because the Hawking evaporation lifetime scales as $\tau_{\text{BH}} \propto G^{-2}$, the suppression of the gravitational coupling at that epoch implies longer PBH lifetimes and hence a higher survival mass threshold than predicted by standard general relativity (GR).

This document derives the PDL prediction for the PBH survival threshold M_{PDL}^* , characterises the resulting observational signal in the isotropic gamma-ray background (IGRB), and identifies the precise test that the community can perform with existing Fermi-LAT data and public spectral codes.

2 The PDL effective gravitational coupling at the PBH epoch

2.1 The coupling suppression factor

The cosmological epoch relevant to PBH formation and early evolution is characterised in PDL by the closure density $N_{\text{CMB}} = 40$, derived from the neutron quintuplet structure alone [6]:

$$N_{\text{CMB}} = \lfloor \Gamma_n \rfloor = \lfloor 40.102 \rfloor = 40, \quad \Gamma_n = 6\mu_n - R_{\text{tot}}(n), \quad (2)$$

where μ_n is the neutron's coherence cycle count and $R_{\text{tot}}(n) = 10992$ is its total relational budget. No cosmological parameter is used as input.

The PDL surface fraction is $\kappa_{\text{PDL}} = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. This is an unconditional theorem of axioms C1–C4 [5]. Numerically:

$$\kappa_{\text{PDL}} = \frac{310\varphi}{11017} \approx 0.045529. \quad (3)$$

The suppression factor at the CMB epoch is therefore:

$$\sigma(40) = 1 - (1 - \kappa_{\text{PDL}})^{40} \approx 0.844935, \quad (4)$$

and the effective gravitational coupling at that epoch is:

$$G_{\text{eff}}(N_{\text{CMB}}) = \sigma(40) \cdot G_{\text{PDL}} \approx 0.8449 G_{\text{Newton}}. \quad (5)$$

2.2 Impact on the Hawking evaporation lifetime

The Hawking evaporation lifetime of a Schwarzschild black hole of mass M is [20, 21]:

$$\tau_{\text{BH}}(M) = \frac{5120 \pi G^2 M^3}{\hbar c^4 f(g^*)}, \quad (6)$$

where $f(g^*) \approx 3.8$ for the Standard Model degrees of freedom $g^* = 106.75$ [10]. Because $\tau_{\text{BH}} \propto G^2$, replacing G by $G_{\text{eff}}(N_{\text{CMB}}) = \sigma(40) \cdot G$ yields:

$$\tau_{\text{PDL}}(M) = \frac{\tau_{\text{GR}}(M)}{\sigma(40)^2}, \quad (7)$$

so PDL lifetimes at the PBH epoch are longer than GR lifetimes by a factor:

$$\frac{\tau_{\text{PDL}}}{\tau_{\text{GR}}} = \sigma(40)^{-2} = 1.4007. \quad (8)$$

3 The PDL survival threshold

3.1 Derivation

The standard GR survival threshold M_{GR}^* is defined by $\tau_{\text{GR}}(M_{\text{GR}}^*) = t_{\text{U}}$, where $t_{\text{U}} \approx 13.8$ Gyr is the age of the universe [22]. The canonical value is [21, 10]:

$$M_{\text{GR}}^* \approx 5.1 \times 10^{14} \text{ g}. \quad (9)$$

The PDL threshold M_{PDL}^* is defined by $\tau_{\text{PDL}}(M_{\text{PDL}}^*) = t_{\text{U}}$. Using equation (7):

$$\frac{\tau_{\text{GR}}(M_{\text{PDL}}^*)}{\sigma(40)^2} = t_{\text{U}} \quad \Rightarrow \quad \tau_{\text{GR}}(M_{\text{PDL}}^*) = \sigma(40)^2 t_{\text{U}}. \quad (10)$$

Since $\tau_{\text{GR}} \propto M^3$, this gives:

$$\frac{M_{\text{PDL}}^*}{M_{\text{GR}}^*} = \left(\frac{t_{\text{U}}}{\sigma(40)^2 t_{\text{U}}} \right)^{1/3} = \sigma(40)^{-2/3}. \quad (11)$$

Falsifiable prediction — PDL survival threshold

$$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 1.11888 M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}. \quad (12)$$

The shift relative to the GR value is +11.89%, derived from a single unconditional theorem (κ_{PDL}) with no free parameter.

3.2 Numerical verification

All numerical values are derived from the exact expression $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$:

Quantity	Value	Source
φ	$(1 + \sqrt{5})/2$	exact
κ_{PDL}	$310\varphi/11017 \approx 0.04552878$	[5]
$\sigma(40)$	$1 - (1 - \kappa_{\text{PDL}})^{40} \approx 0.84493508$	equation (4)
$\sigma(40)^{-2}$	1.400726	
$\sigma(40)^{-2/3}$	1.118882	equation (11)
M_{GR}^*	$5.100 \times 10^{14} \text{ g}$	[21, 10]
M_{PDL}^*	$5.706 \times 10^{14} \text{ g}$	equation (12)
$\Delta M^*/M_{\text{GR}}^*$	+11.89%	

Table 2: Numerical values entering the PDL threshold prediction. All quantities flow from the single unconditional theorem $\kappa_{\text{PDL}} = 310\varphi/11017$ and the derived value $N_{\text{CMB}} = 40$.

Internal consistency check. One may verify the coherence of the result: $(M_{\text{PDL}}^*/M_{\text{GR}}^*)^3 = \sigma(40)^{-2} = 1.400726$, confirming that the modified threshold indeed satisfies $\tau_{\text{PDL}}(M_{\text{PDL}}^*) = t_{\text{U}}$ exactly.

4 The observational signature

4.1 Physical picture

PBHs in the mass window $[M_{\text{GR}}^*, M_{\text{PDL}}^*] = [5.10, 5.71] \times 10^{14} \text{ g}$ present the following behaviour under the two frameworks:

- *Standard GR:* $\tau_{\text{GR}}(M) < t_{\text{U}}$ for all $M < M_{\text{GR}}^*$, so these PBHs have fully evaporated and contribute zero flux to the present-day IGRB.

- *PDL*: $\tau_{\text{PDL}}(M) > t_{\text{U}}$ for all $M > M_{\text{PDL}}^*$. PBHs with $M \in [M_{\text{GR}}^*, M_{\text{PDL}}^*]$ have $\tau_{\text{GR}}(M) < t_{\text{U}}$ yet $\tau_{\text{PDL}}(M) > t_{\text{U}}$, so they are still active today and contribute positively to the IGRB.

The PDL prediction therefore implies an *excess of present-day Hawking radiation flux* from the mass window $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$ relative to the GR baseline.

4.2 Spectral localisation

The Hawking temperature of a black hole of mass M is:

$$T_H(M) = \frac{\hbar c^3}{8\pi G M k_B}, \quad (13)$$

which evaluates to:

$$T_H(M_{\text{GR}}^*) \approx 20.7 \text{ MeV}, \quad (14)$$

$$T_H(M_{\text{PDL}}^*) \approx 18.5 \text{ MeV}. \quad (15)$$

The peak photon energy in the Hawking spectrum scales approximately as $E_{\text{peak}} \sim 5 T_H$ [21], giving:

$$E_{\text{peak}} \approx 93\text{--}104 \text{ MeV}, \quad (16)$$

which falls squarely within the Fermi-LAT energy range of 20 MeV to 300 GeV [23].

4.3 Nature of the signal

The signal is not a shift in the *abundance* of PBHs, which PDL does not predict. It is a shift in the *survival threshold*: the lower spectral cutoff of the Hawking contribution to the IGRB is displaced from $E_{\text{peak}}(M_{\text{GR}}^*) \approx 104 \text{ MeV}$ to $E_{\text{peak}}(M_{\text{PDL}}^*) \approx 93 \text{ MeV}$. In practice, if the initial PBH abundance f_{PBH} is non-zero in the window $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$, one expects:

- An excess of IGRB flux at $E \lesssim 104 \text{ MeV}$ relative to the GR prediction.
- The spectral shape of the excess follows the standard Hawking spectrum [24] evaluated at $T_H \approx 19\text{--}21 \text{ MeV}$.
- The fractional shift in the spectral cutoff is $\Delta E/E \approx 10.6\%$, comparable to the Fermi-LAT spectral resolution at 100 MeV ($\sim 10\text{--}20\%$).

4.4 Additional prediction: entropy suppression

A complementary prediction concerns the integrated entropy-to-energy ratio of PBH-seeded objects at high redshift. The PDL suppression factor at epoch N is $\sigma(N)^2$, so:

$$\frac{S_{\text{PDL}}}{E_{\text{PDL}}} \propto \sigma(N(z))^2 \cdot \left(\frac{S}{E}\right)_{\text{GR}}, \quad (17)$$

predicting a systematic redshift dependence of the entropy-to-energy ratio in AGN populations seeded by PBHs [12]. At z corresponding to $N = 40$ ($z \sim z_{\text{CMB}}$), the suppression factor is $\sigma(40)^2 = 0.7139$, i.e. a 28.6% reduction relative to the local value.

5 How to test this prediction

The PDL prediction can be tested quantitatively using entirely public tools and data.

Step 1 — Compute the Hawking spectra. Use BLACKHAWK [7] (v2.1 or later) to compute the differential photon flux $d\Phi_\gamma/dE$ for a monochromatic PBH mass distribution centred at $M = M_{\text{GR}}^*$ and at $M = M_{\text{PDL}}^* = 5.706 \times 10^{14}$ g separately. The code integrates over all Standard Model degrees of freedom and includes both primary and secondary (fragmentation) contributions.

Step 2 — Normalise to the IGRB constraints. Use ISATIS [8] to compare the computed spectra against the Fermi-LAT IGRB measurement of Ackermann et al. [9] and the constraint compilation of Carr et al. [10]. The relevant quantity is the upper limit on $f_{\text{PBH}}(M)$ in the window $[5.10, 5.71] \times 10^{14}$ g.

Step 3 — Identify the discriminating observable. The discriminating quantity is the position of the spectral cutoff in the 50–200 MeV band of the IGRB. Under GR, this cutoff occurs at $E_c \approx 104$ MeV; under PDL, it is displaced to $E_c \approx 93$ MeV. A fit to the Fermi-LAT IGRB spectrum in this band with E_c as a free parameter constitutes a direct test of the PDL prediction.

Step 4 — Complementary constraint from the entropy-to-energy ratio. An independent test uses the redshift dependence of the entropy-to-energy ratio in AGN populations. The predicted scaling $S/E \propto \sigma(N(z))^2$ can be confronted with multi-epoch surveys such as those from DESI or Euclid.

6 Relation to the PDL causal chain

The PBH threshold prediction is not an isolated result but a corollary of the causal chain established in the PDL corpus. The complete chain from axioms to the present prediction is:

$$\begin{array}{ccccccc} \text{C1–C4} & \xrightarrow{\text{D16a}} & K_4 & \xrightarrow{\text{D16b, D29}} & \text{quintuplet} & \xrightarrow{\text{D39, D42}} & \kappa_{\text{PDL}} \xrightarrow{\text{D24}} \\ \sigma(N) & \xrightarrow{\text{D31, D36, D42}} & G_{\text{eff}}(N) & \xrightarrow{\text{D27}} & N_{\text{CMB}} = 40 & \xrightarrow{\text{D38}} & M_{\text{PDL}}^* \end{array}$$

Every arrow in this chain has been proved as an unconditional theorem of C1–C4 (see D43v3 [3] for the complete synthesis). The prediction M_{PDL}^* inherits the unconditional character of each link; it contains no free parameter beyond the Standard Model degrees of freedom g^* entering the Hawking spectrum, which are taken from the literature [10].

Position within the PDL corpus. This document constitutes D45 of the PDL corpus. The global mapping of all documents, open problems, and resolved results is maintained in the D-Map [25], currently at version 16.

7 Discussion

What this prediction does and does not claim. The PDL framework predicts the *position* of the PBH survival threshold, not the *abundance* of PBHs in any mass window. The observational signal therefore depends on the initial PBH mass function, which is cosmological model-dependent and not determined by PDL. If f_{PBH} is negligibly small in $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$, the signal will be below any present detection threshold. In that case the prediction cannot be confirmed, but it is also not falsified. Falsification would require demonstrating that the IGRB spectral cutoff is definitively located at $E_c^{\text{GR}} \approx 104 \text{ MeV}$ rather than at $E_c^{\text{PDL}} \approx 93 \text{ MeV}$, together with a non-negligible f_{PBH} in the relevant window.

Comparison with other modified-gravity scenarios. Several alternative frameworks also predict modifications to the Hawking evaporation rate, including models with extra dimensions [26], memory-burden effects [27], and dark charges [28]. The PDL prediction is distinguished by three features: (i) the shift is derived from a single unconditional combinatorial theorem ($\kappa_{\text{PDL}} \in \mathbb{Q}(\sqrt{5})$), not from a new adjustable parameter; (ii) the same framework simultaneously derives G , α , μ^* , and the Hubble tension resolution; (iii) the shift is positive (heavier threshold) and of a specific magnitude (+11.89%) that does not depend on any model tuning.

Sensitivity requirements. The spectral displacement $\Delta E/E \approx 10.6\%$ is at the boundary of current Fermi-LAT sensitivity at 100 MeV. A dedicated analysis using the full 15+ years of Fermi-LAT data, focusing on the IGRB spectrum in the 50–200 MeV window, would provide the most constraining test. Next-generation MeV telescopes such as AMEGO-X [29] and e-ASTROGAM [30] would improve the sensitivity substantially, potentially resolving the spectral cutoff position to better than 5%.

8 Conclusion

The PDL framework predicts that the primordial black hole survival mass threshold is shifted upward by +11.89% relative to the standard GR value, giving:

$$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}. \quad (18)$$

This prediction is an unconditional corollary of the density-dependent effective gravitational coupling $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, itself proved from the PDL axioms C1–C4 without any free parameter. The resulting observable is a shift of ≈ 11 MeV in the spectral cutoff of the Hawking contribution to the isotropic gamma-ray background, in the ~ 90 –105 MeV range accessible to Fermi-LAT.

The test requires computing the Hawking photon spectrum with BLACKHAWK at $M = 5.706 \times 10^{14} \text{ g}$ and comparing it with the Fermi-LAT IGRB data using ISATIS. This is a standard numerical procedure well within the capacity of existing codes and data. We invite the community to perform this test and report the result, whether confirmatory or falsifying.

Falsifiable prediction — Summary

PDL prediction D45: The PBH survival mass threshold satisfies $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 5.706 \times 10^{14} \text{ g}$. The Hawking flux from PBHs in the window $[5.10, 5.71] \times 10^{14} \text{ g}$ contributes a non-zero signal to the present-day IGRB at $E_{\text{peak}} \approx 93$ –104 MeV. This signal is absent in standard GR. It can be tested with BLACKHAWK + ISATIS + Fermi-LAT IGRB data.

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